

Lecture 4: Art of Approximation

Doing More with Less

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This lecture covers few topics around approximation techniques. Interestingly, these topics are not covered in a lot of the courses, which I found it is extremely helpful on job and dealing with interview questions. Toward the end of the lesson, students should be able to recognize patterns that could apply approximation techniques in solving certain math problems quickly.

Please do not hesitate to reach me via WhatsApp, either in the group or PM, if you have any questions regarding the course.

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1 Dimensional Analysis

1.1 Physical Problem

Imagine we never learn any basic physics - we don't even know the relationship between speed (v), distance(D) and time(t). How can we deduce the relationship between them? Intuitively, it should be $v = \frac{D}{t}$. Have you ever wonder how you arrive this conclusion so quickly?

Let's look at the problem from another angle.

What is the dimension of each quantity?

- Speed (v): Meter per second ($[\frac{m}{s}]$)
- Distance (D): Meter ($[m]$)
- Time (t): Second ($[s]$)

The next question is, what can we represent speed with the remaining two variables: distance and time? What about $v = \frac{D}{t^2}$? What about $v = \frac{D^2}{t^3}$?

Both are not correct indeed. Why? Let's look at their dimension:

- $v = \frac{D}{t^2} \rightarrow \frac{m}{s^2} \neq \frac{m}{s}$
- $v = \frac{D^2}{t^3} \rightarrow \frac{m^2}{s^3} \neq \frac{m}{s}$

Their dimensions do not even align! The core message here is *"You can't compare apple to orange"*. This **common sense** is a very powerful tool that can be applied to solve more complicated problem.

1.2 Mathematical Problem

Look at this integration:

$$\int_{-\infty}^{\infty} e^{-\alpha x^2} dx \quad (1)$$

Assume x takes a dimension of length (L), we may apply similar technique in the previous section to guesstimate the integral. When we do the integration, the x is removed by the integral, therefore the only variable that should remain is α .

$$\int_{-\infty}^{\infty} e^{-\alpha x^2} dx = f(\alpha) \quad (2)$$

Let's observe the dimension of the integral.

- 1) Exponent needs to be dimensionless,
- 2) e is dimensionless, therefore $e^{-\alpha x^2}$ is dimensionless,
- 3) dx has a dimension of length (L)

With these information, we may deduce that,

- From 1) and 2), $[\alpha] = m^{-2}$

- From 3), $[\int_{-\infty}^{\infty} e^{-\alpha x^2} dx] = [dx] = m$
- Therefore, $\int_{-\infty}^{\infty} e^{-\alpha x^2} dx \propto \frac{1}{\sqrt{\alpha}}$

Now we have the relationship $\int_{-\infty}^{\infty} e^{-\alpha x^2} dx \propto \frac{1}{\sqrt{\alpha}}$, which translates to:

$$\int_{-\infty}^{\infty} e^{-\alpha x^2} dx = C \times \frac{1}{\sqrt{\alpha}} \quad (3)$$

for a constant C . To figure out C , it requires knowledge of Gaussian integral:

$$\int_{-\infty}^{\infty} e^{-x^2} dx = \sqrt{\pi} \quad (4)$$

By comparing the two equations, we may figure out that $C = \sqrt{\pi}$.

2 Extreme Case Analysis

2.1 More on Common Sense

Instead of asking you to do the integral using dimensional analysis, now let's answer this question in 5 seconds:

$$\int_{-\infty}^{\infty} e^{-\alpha x^2} dx = \begin{cases} (a) \sqrt{\alpha\pi} & (?) \\ (b) \sqrt{\frac{\pi}{\alpha}} & (?) \end{cases} \quad (5)$$

It should be (a). Why? Let's look at an extreme case $\alpha \rightarrow \infty$.

$$\alpha \rightarrow \infty \Rightarrow e^{-\alpha x^2} \rightarrow 0 \Rightarrow \int_{-\infty}^{\infty} e^{-\alpha x^2} dx = 0 \quad (6)$$

Which option will lead to this result? It is (b).

You may repeat this analysis using another extreme case to arrive the same conclusion: $\alpha \rightarrow 0$.

$$\alpha \rightarrow 0 \Rightarrow e^{-\alpha x^2} \rightarrow 1 \Rightarrow \int_{-\infty}^{\infty} e^{-\alpha x^2} dx = \int_{-\infty}^{\infty} 1 dx = \infty \quad (7)$$

2.2 Geometry Example

Now, we may look at a more visual example to illustrate this concept. You may like to know what is the formula to compute the area of the ellipse. Here are five possible options.

$$Area = \begin{cases} (a) & ab^2, \\ (b) & a^2 + b^2, \\ (c) & \frac{a^3}{b}, \\ (d) & 2ab \\ (e) & \pi ab \end{cases} \quad (8)$$

By applying dimensional analysis, you may rule out (a). How may we apply extreme case analysis

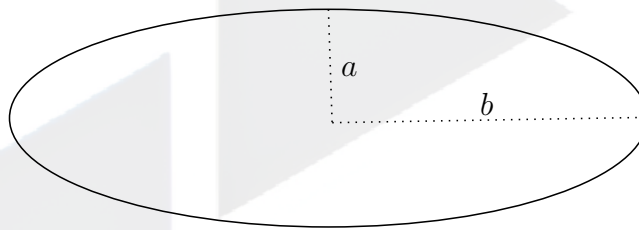


Figure 1: A generic ellipse

to rule out more?

Imagine now one of the axis becomes 0. The area of the ellipse will also goes to 0 regardless of the length of another axis. Therefore, we may rule out (b) and (c).

Finally, what about $a = b$? It resembles the simplest case - a circle, which it's area is equal to πr^2 . To fulfill this, the only choice that will recover this formula is (e) - another "extreme" case analysis.

2.3 Physical Example

This technique is actually a very common technique in physic. By understanding how it is applied to a physical system, you may start to build intuition on applying this technique to any real world system, for instance, financial modeling.

One of the most fundamental physics problem is calculating the period of oscillation given the following variables:

- Mass (M - Dimension kg) of the ball
- Spring constant (k - Dimension $[\frac{kg}{s^2}]$) - Measuring the "hardness" of the spring

Here is the formula of the period:

$$T = 2\pi \sqrt{\frac{M}{k}} \quad (9)$$

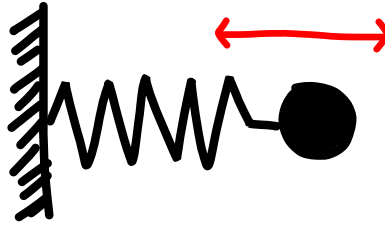


Figure 2: A physical system consist of a ball attached to a spring

If we never saw this equation before, how can we verify if the equation is making sense or not? By plugging in extreme values like $M \rightarrow \infty$ or $k \rightarrow \infty$, one may immediate check the result against "common sense" (i.e. sanity check) to the equation.

2.4 Finance Example

A simple example is present value calculation. To calculated the present value of a future cash-flow, one needs to discount the future received value with a discount factor corresponding to the interest rate:

$$PV = \frac{C}{(1+i)^n} \quad (10)$$

You may ask, does this equation making sense? Let's look at two extreme case.

Let's say the interest rate is ∞ (i.e. $i \rightarrow \infty$). Intuitively, any future cashflow worth nothing as if you put \$1 in bank, it will become $\$ \infty$ in just a second. (10) aligns with this understanding.

For another extreme case, let's assume there is a future cashflow receive after ∞ years (i.e. $n \rightarrow \infty$). In this case, the future cashflow worth nothing since I am not going to receive it within finite amount of time anyway. It also aligns with (10).

Market Impact Model

In high frequency trading, one often needs to estimate the price impact of a trade. There of numbers of models out there trying to estimate the price impact of any trade order. There is a famous model - squared root law, that is commonly used in firms for estimating the market impact of a trade order. Here is the equation:

$$P = c \cdot \sigma \cdot \sqrt{\frac{n}{v}} \quad (11)$$

with,

- P: Market Impact
- c: Constant
- σ : Volatility

- n: Order size
- v: Traded volume

Question: Does it make sense? Can you analyze it with extreme case analysis?

3 Numerical Approximations

In day to day work, especially traders or quants, often time one needs to get some approximation quickly to make decisions. That is one of the reason why these types of approximation questions are commonly seen during the interview. This section will covers different techniques in doing numerical approximation.

3.1 Multiplication

Here is a simple example to illustrate the idea. Suppose you would like to estimate the 31.5×721 , the easiest estimation is by only retaining the first digit and do the multiplication: $31.5 \times 721 \approx 30 \times 700 = 21000$.

If we expand the calculation like the following,

$$31.5 \times 721 = (30 + 1.5) \times (700 + 21) = 30 \times (1 + 0.05) \times 700 \times (1 + 0.03)$$

By reordering the formulate,

$$30 \times 700 \times (1 + 0.05) \times (1 + 0.03)$$

For the later part of the product, $(1 + 0.05) \times (1 + 0.03)$ is very close to one. In case we would like to improve the accuracy,

$$(1 + 0.05) \times (1 + 0.03) = 1 \times 1 + 0.05 \times 1 + 1 \times 0.03 + 0.05 \times 0.03 \approx 1 + 0.05 + 0.03 = 1.08$$

With this correction, we may improve the first approximation,

$$31.5 \times 721 \approx 21000 \times 1.08 = 22680$$

Now moving on to a generic case, given two numbers: $(x + \Delta x)$ and $(y + \Delta y)$.

$$(x + \Delta x)(y + \Delta y) = xy + x\Delta y + y\Delta x + \Delta x \Delta y$$

After some rearrangement,

$$(x + \Delta x)(y + \Delta y) = xy \times \left(1 + \frac{\Delta x}{x}\right) \left(1 + \frac{\Delta y}{y}\right) = xy \times \left(1 + \frac{\Delta x}{x} + \frac{\Delta y}{y} + \frac{\Delta x}{x} \frac{\Delta y}{y}\right)$$

Similar technique can also be applied to square of numbers,

$$(x + \Delta x)^2 = x^2 \times \left(1 + \frac{\Delta x}{x}\right)^2 \approx x^2 \times \left(1 + 2 \cdot \frac{\Delta x}{x}\right)$$

3.2 Reciprocals

For reciprocals, here is an interesting fact,

$$\frac{1}{1+x} \approx 1-x \quad \text{if } x \rightarrow 0$$

How can this be applied? For instance, you would like to estimate $\frac{1}{13}$.

$$\frac{1}{13} = \frac{1}{10} \times \frac{1}{1+0.3} \approx \frac{1}{10} \times 0.7 = 0.07$$

which is very close to the actual value 0.0769.

3.3 Square Root

For square root, one may also do similar approximations. Given we want to calculate $\sqrt{10}$.

$$\sqrt{10} = \sqrt{9} \times \sqrt{1 + \frac{1}{9}}$$

Here is a fact,

$$\sqrt{1+x} \approx 1 + \frac{x}{2} \quad \text{when } x \rightarrow 0$$

Therefore,

$$\sqrt{10} = \sqrt{9} \times \sqrt{1 + \frac{1}{9}} \approx 3 \times \left(1 + \frac{1}{18}\right) = 3\frac{1}{6} = 3.1666\dots$$

where the actual value is 3.1623. It is very close!

3.4 General Case

For those coming from a mathematical or engineering background would realize all above approximations are based on a simple technique: Taylor expansions.

$$(1+x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots + \frac{n(n-1)\dots(n-i+1)}{i!}x^i + o(x^i)$$

If x is small, then the later terms can be omitted.

$$(1+x)^n \approx 1 + nx$$

By substituting n with $1/2$, -1 , and other numbers, you may deduce different approximation methods for quick numerical computations.